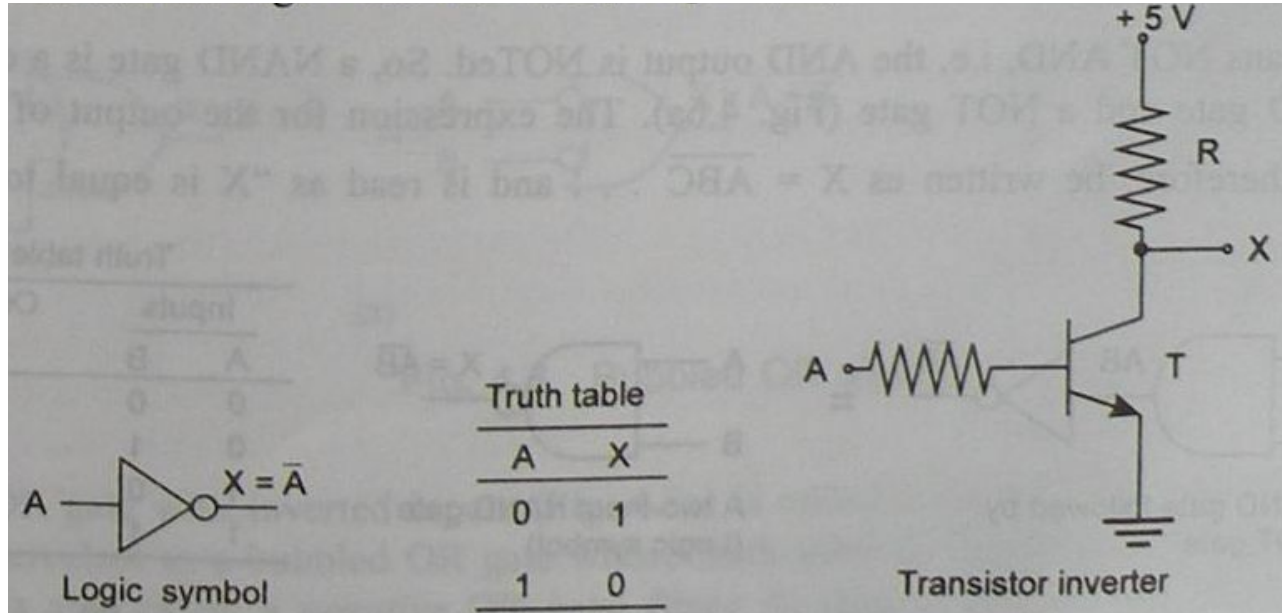


Logic Gates and Boolean Algebra

LOGIC GATES

- Fundamental building blocks of digital systems
- Ability of decision making
- ***Electronic circuits*** because made up of electronic devices and components
- I/P and O/P occur only in 2 levels: HIGH and LOW (or) TRUE and FALSE (or) ON and OFF (or) 1 and 0
- **Truth Table:** Lists all possible combinations of input variables and the corresponding outputs
- **Positive logic system:** Higher of the two voltage levels represents the logic 1 and lower voltage level represents logic 0
- **Negative logic system:** Lower of the two voltage levels represents the logic 1 and higher voltage level represents logic 0

NOT GATE (or) INVERTER



Buffer

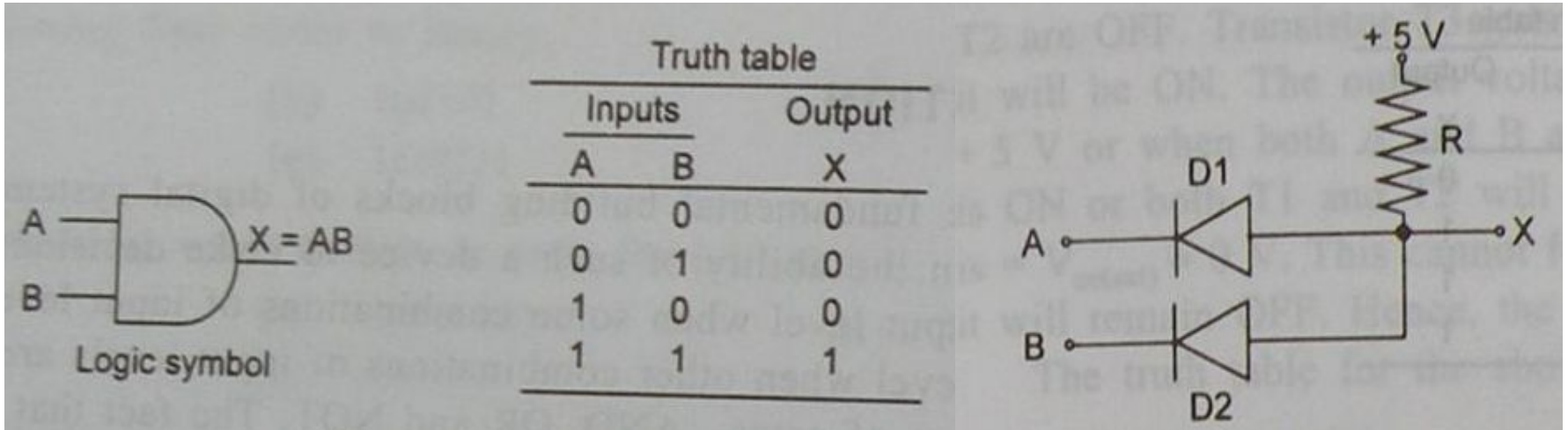


$$F = x$$

x	F
0	0
1	1

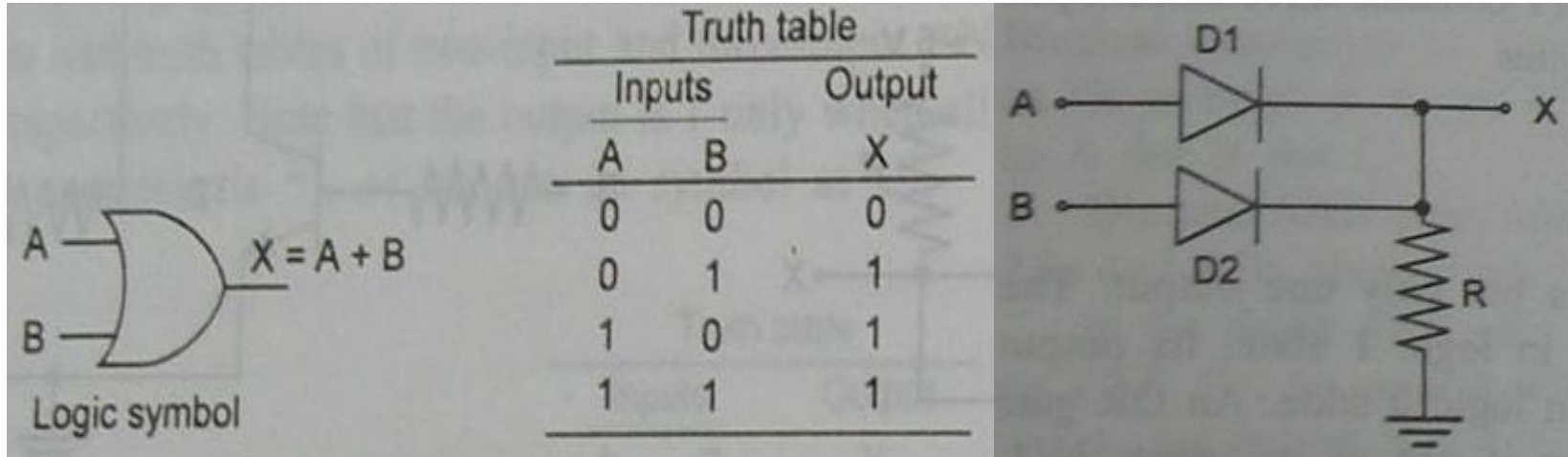
AND GATE

Also known as **All** or **Nothing Gate**



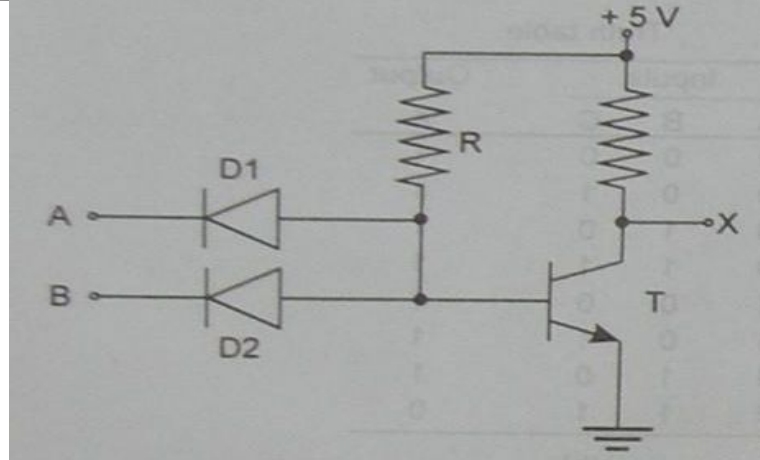
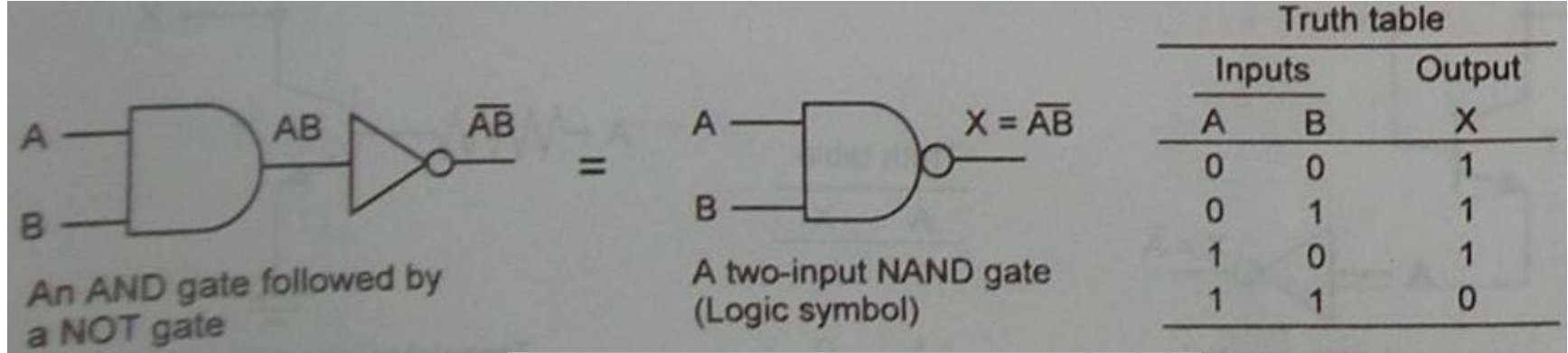
OR GATE

Also known as **Any** or **All Gate**

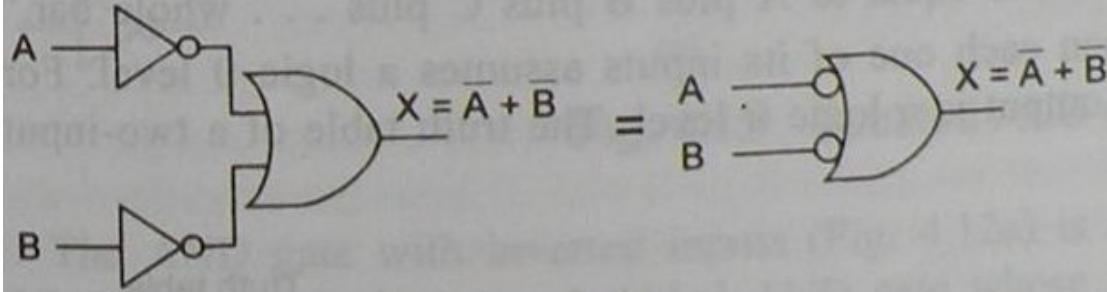


- Logic circuits which use these three gates only are called **AND/OR/INVERT**, i.e. **AOI** logic circuits.
- Logic circuits which use only AND and OR gates are known as **AND/OR**, i.e. **AO** logic circuits.

NAND GATE (UNIVERSAL GATE)



$$X = \overline{AB} = \overline{A} + \overline{B}$$



(a)

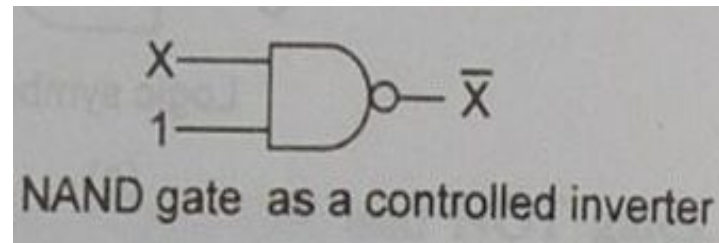
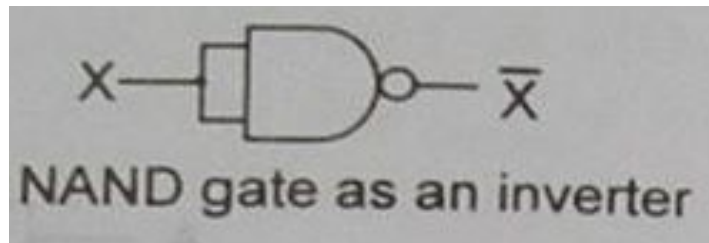
Truth table

Inputs		Inverted Inputs		Output
A	B	$\overline{A}$	$\overline{B}$	X
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

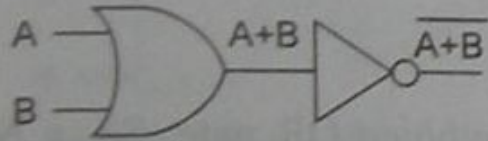
(b)

OR gate with inverted inputs: **Bubbled OR gate** (or) **Negative OR gate**

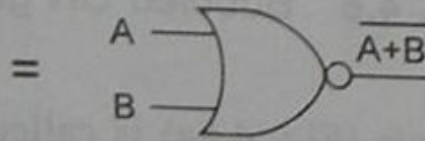
NAND gate: **Active Low OR gate**



NOR GATE (UNIVERSAL GATE)



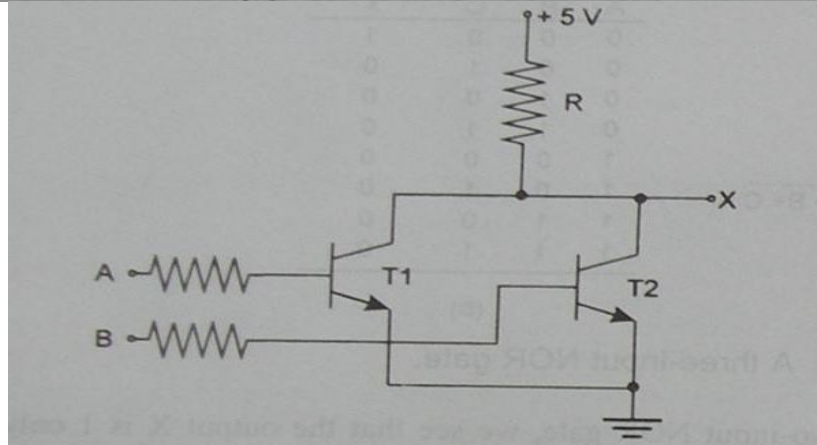
An OR gate followed by a NOT gate



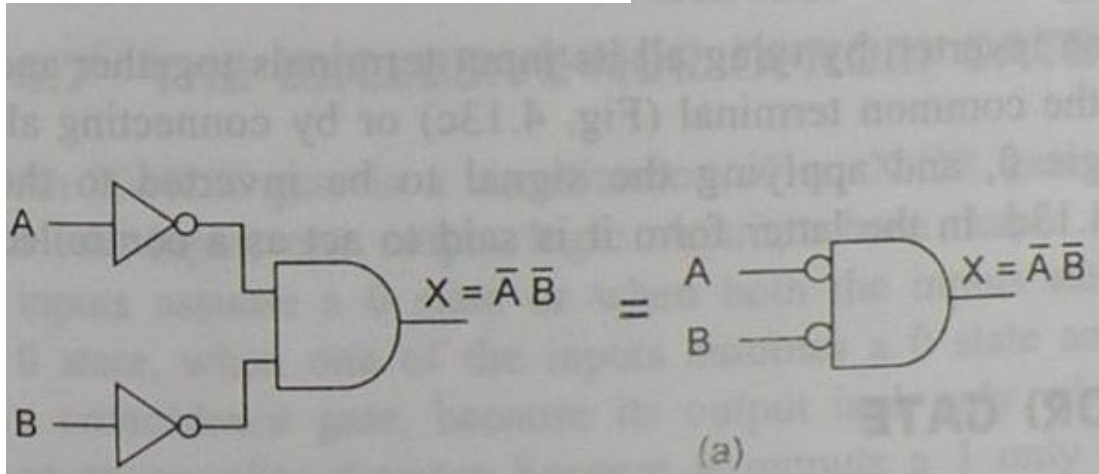
A two-input NOR gate (logic symbol)

Truth table

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0



$$X = \overline{A + B} = \overline{A} \overline{B}$$



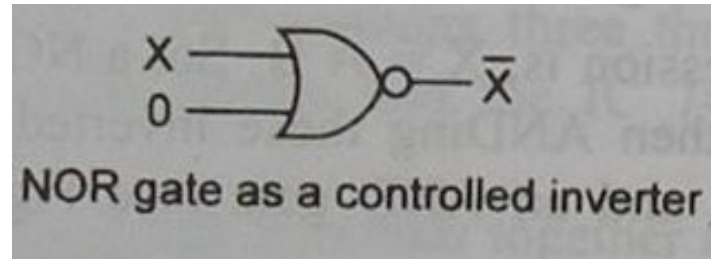
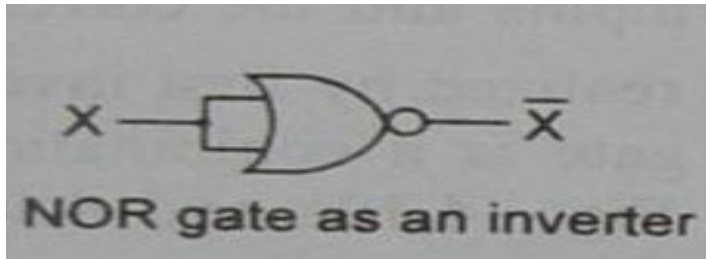
Truth table

Inputs		Inverted inputs		Output
A	B	$\overline{A}$	$\overline{B}$	X
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

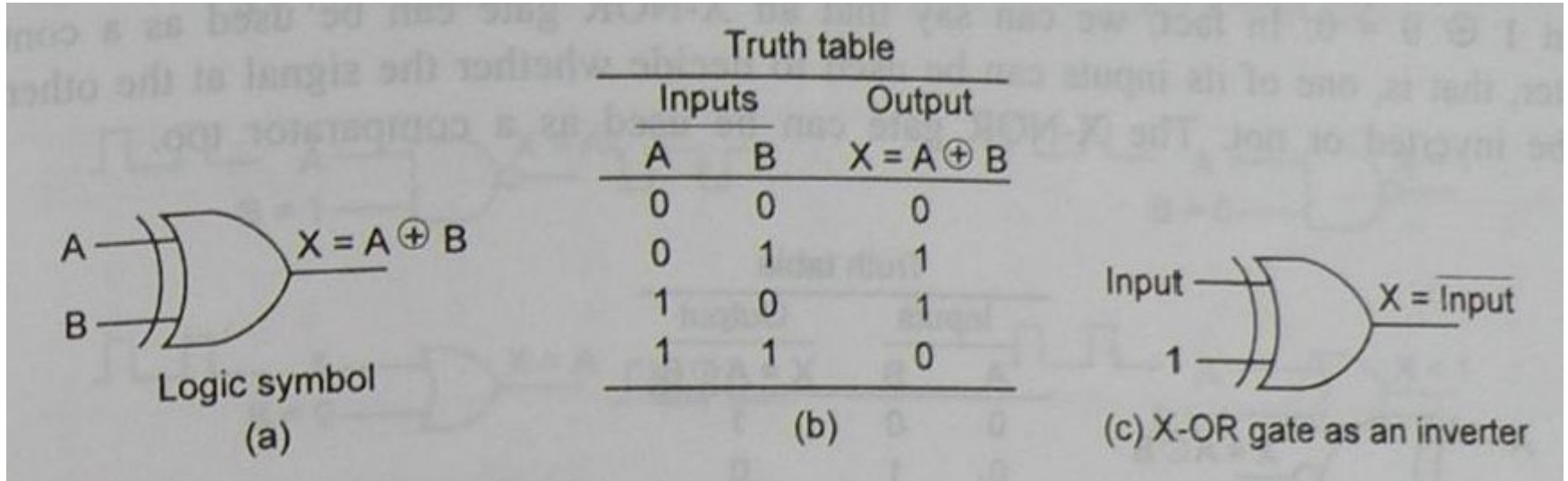
(b)

AND gate with inverted inputs: **Bubbled AND gate** (or) **Negative AND gate**

NOR gate: **Active Low AND gate**



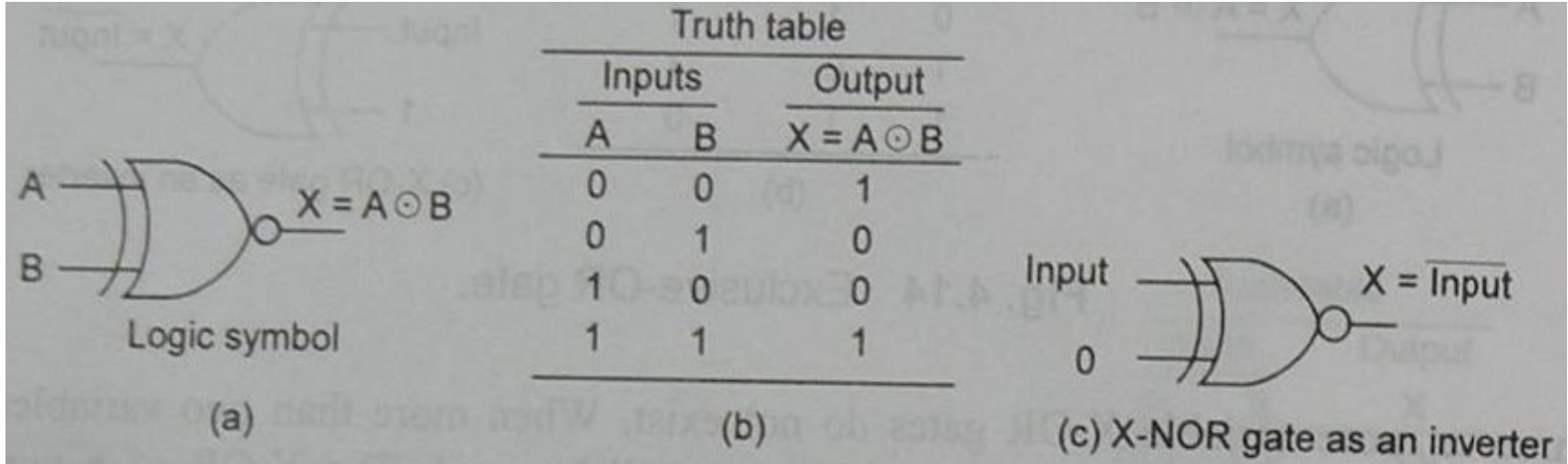
EXCLUSIVE-OR (X-OR) GATE



$$X = A \oplus B = A\bar{B} + \bar{A}B$$

- **Anti-Coincidence gate** (or) **Inequality detector**: Produces O/P 1 only when inputs are not equal
- 3 or more variables X-OR gates are not available

EXCLUSIVE-NOR (X-NOR) GATE



$$X = A \odot B = AB + \bar{A}\bar{B} = \overline{A \oplus B} = \overline{A\bar{B} + \bar{A}B}$$

- **Coincidence gate** (or) **Equality detector**: Produces O/P 1 only when inputs are equal
- 3 or more variables XNOR gates are not available

The X-NOR of two variables A and B is the complement of the X-OR of those two variables. That is,

$$A \odot B = \overline{A \oplus B}$$

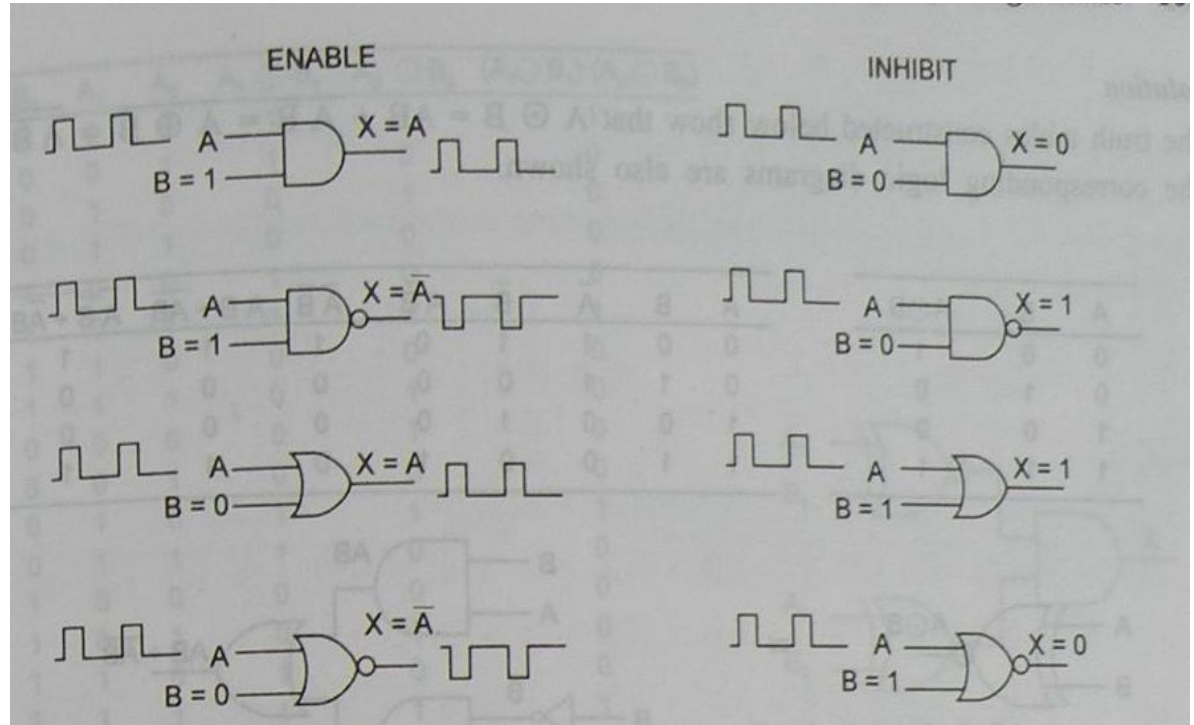
But the X-NOR of three variables A, B and C is not equal to the complement of the X-OR of A, B, and C. That is,

$$A \odot B \odot C \neq \overline{A \oplus B \oplus C}$$

However, the X-NOR of a number of variables is equal to the complement of the X-OR of those variables only when the number of variables involved is even.

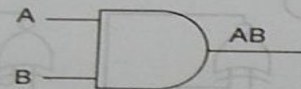
INHIBIT GATE

- I/P is not able to reach the O/P
- O/P is constantly low or high

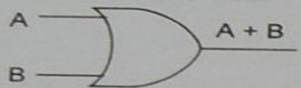


Traditional symbol

(a) AND Gate



(b) OR Gate



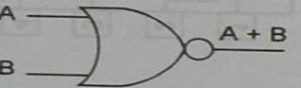
(c) NOT Gate



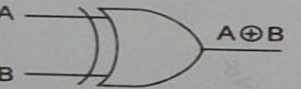
(d) NAND Gate



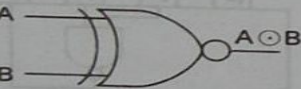
(e) NOR Gate



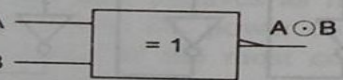
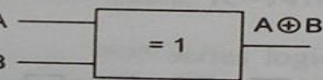
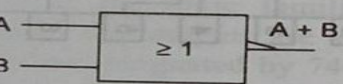
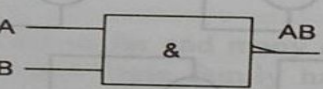
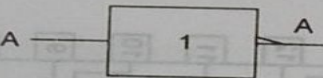
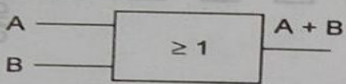
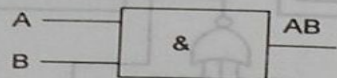
(f) X-OR Gate



(g) X-NOR Gate



ANSI/IEEE symbol



or

